

BINH GIA NGUYEN

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Google Scholar ◊ GitHub ◊ LinkedIn

EDUCATION

FPT University

B.Sc. in Artificial Intelligence

GPA: 8.59/10 Graduated with Honors

Thesis: *LiteVLA-MS: Resolving Motion Under-Specification in Lightweight Vision-Language-Action Models*

2022 – 2026

Hanoi, Vietnam

EXPERIENCES

Center for AI Research (CAIR), VinUniversity · Research resident

Apr 2025 – Present

Upwork · AI Freelancer · Remote

May 2024 – Apr 2025

BSM Labs · AI Engineer · Hanoi, Vietnam

Jun 2024 – Feb 2025

Aimesoft · AI Intern · Hanoi, Vietnam

Mar 2024 – Jun 2024

RESEARCH INTERESTS

I am interested in **Efficient AI** - specifically model compression, model pruning and sparsity, efficient training and inference acceleration, and scalable deployment of large-scale models.

RESEARCH & PUBLICATIONS

CKA-Guided Layer Pruning for VLA Models

2025 – Present

Supervisor: Duy H. M. Nguyen

VinUniversity / VinRobotics, Hanoi

Finetuning Vision-Language-Action Models Requires Fewer Layers Than You Think

Binh G. Nguyen*, Trong-Bao Ho, Thien-Loc Ha, Khoa Vo, Philip Lund Møller, Quang Tan Nguyen, Long Dinh, Tuan Quan Dam, Vu N. Duong, Tung Minh Luu, Trung Le, Tran Nguyen Le, Minh Nhat Vu, An Thai Le, Ngan Le, Daniel Sonntag, James Zou, Jan Peters, Duy Minh Nguyen Ho, Vien Anh Ngo

[RSS Workshop 2026](#) · clpvla.github.io

- Found that current state-of-the-art VLA models exhibit significant layer redundancy, which can be removed without significant performance drop
- Proposed CKA-Guided Layer Pruning (CLP), a training-free structural compression method for VLA models using Centered Kernel Alignment (CKA) layer similarity scores.
- Applied CLP to π_0 , GR00T-N1.5, and SmolVLA; evaluated on LIBERO, RoboCasa, and SimplerEnv benchmarks and 10 real-world tasks with significant training time reduction, inference speedup and minimal performance drop.

Future-Oriented Conditioning for VLA Adaptation

2025 – 2026

Supervisor: Duy H. M. Nguyen

VinUniversity / VinRobotics, Hanoi

FOCA: Future-Oriented Conditioning for Data-Efficient Vision-Language-Action Adaptation

Duc M. Nguyen*, Nghiem T. Diep*, **Binh G. Nguyen***, Trong-Bao Ho, Doanh Le, Tan Nguyen, Thien-Loc Ha, Tran Nhiem, Bao Thach, Nhat Tran, Tuan Anh Tran, Artur Habuda, Philip Lund Moeller, Tran Nguyen Le, Daniel Sonntag, Mathias Niepert, Khoa Doan, Vu Duong, Hung Ngo, Minh Vu, Duy MH Nguyen, An Thai Le, Vien Ngo

[ICML 2026](#) · focavla.github.io

- Co-developed FOCA, a data-efficient adaptation method for VLA models via future-oriented action conditioning (co-first author).

Start Right, Arrive Right: Asynchronous Execution via Initial Noise Selection

Trong-Bao Ho*, Quang Tan Nguyen*, Thien-Loc Ha*, **Binh G. Nguyen**, Viet-Thanh Nguyen, Long Dinh, Minh Nhat Vu, Duy Minh Ho Nguyen, An Thai Le, Vien Anh Ngo

[CoRL 2026](#) · [paint-action-chunking.github.io](#)

EquiVLA: A General Framework for Rotationally Equivariant Vision-Language-Action Models

Thien-Loc Ha*, Quang Tan Nguyen*, Trong-Bao Ho*, Long Dinh, Minh Duc Nguyen, **Binh G. Nguyen**, Pham Tri Quang, Minh Nhat Vu, Duy Minh Ho Nguyen, An Thai Le, Vien Anh Ngo

[RSS Workshop 2026](#) · [equivla.github.io](#)

Progress-Guided Preference Generation for Aligning Generative Control Policies

Tung Minh Luu*, Tri Ton, Minh Duc Nguyen, Thanh Xuan Nguyen, **Nguyen Gia Binh**, Quang Tan Nguyen, Vu N. Duong, Daniel Sonntag, Duy Minh Ho Nguyen, Vien Anh Ngo, Chang D. Yoo

[CoRL 2026](#)

Self-Improving VLA Policies: Selected Diffusion Noise for Spurious-Robust Action Smoothing

Minh Duc Nguyen*, Bao-Ngoc Dao, Tung Minh Luu, **Nguyen Gia Binh**, Vinh Tong, Anji Liu, Vu N. Duong, Dung D. Le, Daniel Sonntag, Trung Le, Ngan Le, Jan Peters, An Thai Le, Minh Nhat Vu, Mathias Niepert, Khoa D. Doan, Duy Minh Ho Nguyen, Vien Anh Ngo

Under Review

OPEN-SOURCE CONTRIBUTIONS

Tavish9/any4lerobot

2025

A curated toolbox of data-conversion utilities for the LeRobot framework.
Contributed the RoboCasa → LeRobot dataset conversion pipeline.

HONORS & AWARDS

Champion — Funix GenAI Hackathon (Top 1 / 22 teams)

2024

MindGPT: AI-powered mindmap generation web application.

100% Full Tuition Scholarship

2022 – 2026

FPT University — Highest Math entrance score (86/90).

Honorable Student Award

2022 – 2026

FPT University — Awarded every semester throughout undergraduate study.

SKILLS

Programming	Python, C++
Frameworks	PyTorch, HuggingFace Transformers, LeRobot, LangChain, Triton
Tools	Git, Docker, Weights & Biases, Linux, L ^A T _E X
Research Interest	Efficient AI